

ESTRATTO

ISTITUTO ITALIANO PER L'AFRICA E L'ORIENTE

ISTITUTO UNIVERSITARIO ORIENTALE

Dipartimento di Studi Asiatici

SOUTH ASIAN ARCHAEOLOGY 1997

*Proceedings of the
Fourteenth International Conference of the European
Association of South Asian Archaeologists,
held in the Istituto Italiano per l'Africa e l'Oriente,
Palazzo Brancaccio, Rome, 7-14 July 1997*

edited by

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Volume I



R O M E

ISTITUTO ITALIANO PER L'AFRICA E L'ORIENTE

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**The 'Tiny Steatite Seals' (Incised Steatite Tablets) of Harappa
Some Observations on Their Context and Dating**

Introduction

In the last four editions of *South Asian Archaeology*, we have given accounts of the different seasons of excavation at Harappa from 1989 to 1995 (4th season: Dales & Kenoyer 1992; 5th season: Kenoyer 1993; 6th season: Meadow & Kenoyer 1994; 7th and 8th seasons: Meadow & Kenoyer 1997; see also Meadow, ed. 1991: 1st through 5th seasons). In this edition we continue the tradition for the 9th and 10th seasons but focus on two specific areas of the site – the north end of Mound AB test trenched in 1996 (Kenoyer & Meadow, this volume) and the eastern margin of Mound E excavated since the 1993 season (this paper). Only partially covered in these reports is a particularly significant aspect of the work of the Harappa Archaeological Research Project (HARP) carried out during the 9th and 10th seasons. This involves an effort to re-investigate previously excavated parts of Harappa (Vats 1940; Wheeler 1947). Thus in 1996, a nearly 10 metre wide (N-S) portion of the west section of Trench I in Mound F was cut back up to five metres and taken to the water table (HARP Trench 37, see below and end note 1). In addition, a remnant of Late Harappan (Cemetery H period) deposit left by Rai Bahadur Daya Ram Sahni just West of Naugauza on Mound AB was carefully removed (HARP Trench 38). In 1997, four small trenches in the area of the 'Granary' were cleared (HARP Trench 41 NE, SE, SW, NW) and the great cutting (HP XXX) into the west side of Mound AB that had been supervised by Margaret Wheeler was cleaned out and the section trimmed back on the north side (HARP Trench 42). (See site map in Kenoyer & Meadow, this volume.)

The purpose of these selected probings into Mounds AB and F, still ongoing, is to tie the earlier work of Sahni, Vats, and Wheeler to that of the Harappa Archaeological Research Project through examination of artifacts from carefully delineated contexts and through a program of dating carbon recovered from hearths and other secure deposits. Already three radiocarbon determinations have justified this approach (Table 1). A date of c. 1740-

1700 BC (calibrated) from Mound AB is the first from Harappa for a well-defined Late Harappan Phase context. The hearth from which it comes, however, belongs to the upper Late Harappan (Period 5) stratum in this area. Thus additional dates are necessary to determine whether the Period 5 occupation at Harappa actually drew to a close well before the middle of the second millennium or continued later in this or other areas.

The second and third dates currently available from our testing of previously excavated zones are c. 2325 BC from a hearth just above the water table in Mound F, Trench I – the ‘deep digging’ of Vats (1940) and c. 2275-2200 from street deposits about 1.5 m above. A brief review of this work is appropriate as an introduction to the excavations half a kilometre away on the east side of Mound E.

TABLE 1 – *Three new radiocarbon dates from Harappa (results quoted using 5568 half-life before AD 1950; calibrations use the University of Washington Quaternary Isotope Laboratory Calibration Program 1993, rev. 3.0.3c)*

Lot [Feat.]	Context	Period	Lab No.	Results	Calibrated (BC $\pm$ 1 sigma)
H96/7313 [18]	Trench 38 Hearth 43 upper fill	Late Harappan (Period 5)	BETA- 93757	3430 $\pm$ 110 bp	1882 (1737, 1714, 1701) 1539
H96/7291 [100]	Trench 37 Hearth 101	Harappan (Period 3B)	BETA- 93756	3870 $\pm$ 50 bp	2455 (2328) 2207
H96/7262 [70]	Trench 37 Street Deposits	Harappa (late Period 3B)	WG (NEC)- 2525	3810 $\pm$ 60 bp	2329 (2272, 2258, 2204) 2140

Mound F, Vats Trench I (HARP Trench 37)

A distinctive feature of Harappa are ‘tiny steatite seals’ and shaped faience and terracotta ‘sealings’ – what we now call ‘incised (steatite) tablets’ and ‘moulded (terracotta and faience) tablets’ (following Joshi & Parpola 1987; Shah & Parpola 1991; Parpola 1994). To date, steatite tablets have been reported only from Harappa. Based on the excavations of M.S. Vats

on Mound F in Trench I, they have long been considered as among the earliest objects specifically shaped to bear the Indus script (c.g., Vats 1940: 324; Parpola 1994: 54, 64, 107). In order to test this presumption, to obtain well stratified collections of sherds and bioarchaeological remains, and to date the deposits, we cut back the west section at the north end of Trench I in the area where Vats had dug below cultural deposits to the water table ('the deep digging') (Fig. 1).

The results of this work confirmed the observations of Vats (1940: 80-110) with respect to the nature of the deposits and to the overall sequence of deposition in this area (Table 2). In the southern half of the ten-metre wide (N-S) cutting, our team encountered compact sediments like those that Vats (1940: 96-97) commented upon, and we found, as he did, that this 'compressed layer gradually becomes deeper green in colour as its

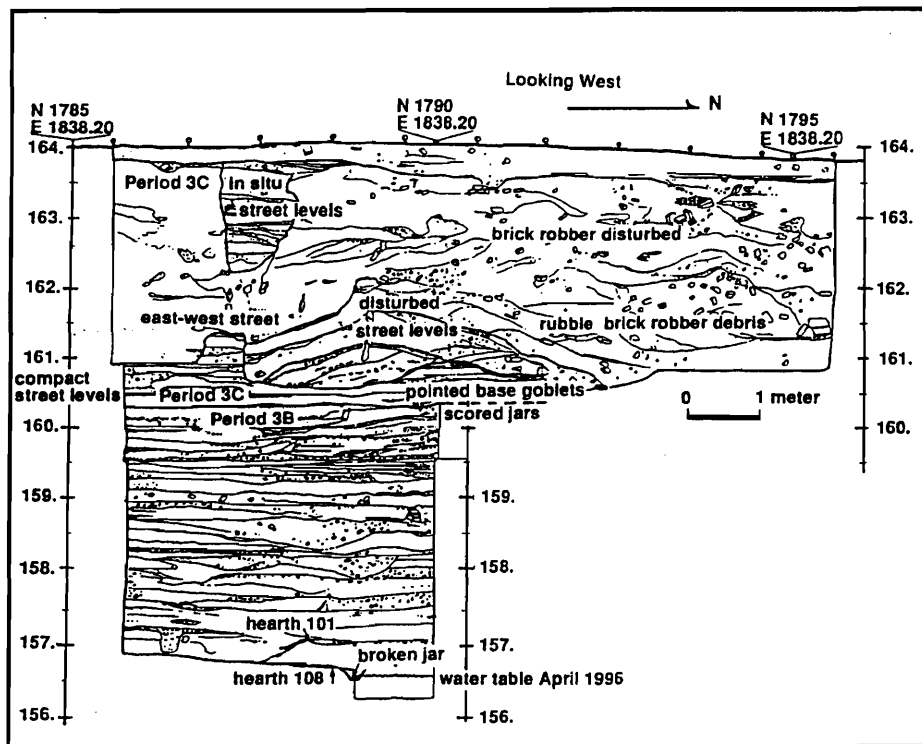


Fig. 1 – Harappa 1996, Mound F, HARP Trench 37 (Vats Trench I): Section

depth increases' (Vats 1940: 97). From our experience elsewhere at Harappa, we know that such deposits are particularly characteristic of streets and that they contain among the best stratigraphic sequences at the site (e.g., Meadow 1991). As Vats also found, brick-robber disturbance had affected even the street deposits to a depth of almost four metres from the surface. Below this there was only a small amount of intact deposit yielding the scored pointed-base (Indus) goblets that are the hallmark of our Period 3C. As excavations in the street continued below about 160.5 m a.m.s.l. (Table 2), we recovered scored goblets without pointed bases in a few excavation units, but lower they too disappeared. Based on analysis of these and other ceramics, most of these lower deposits can be assigned to our Period 3B, which we estimate to date from c. 2450 to c. 2200 BC (e.g., Meadow & Kenoyer 1997). Correlating our elevations with the depth measurements of Vats (Table 2), what we define as Period 3B would lie below about 12 feet and Period 3C material at about 12 feet and above. The lowest pointed-base goblet noted by Vats (1940: 95, No. 39) comes from 12 feet below surface (in upper Stratum V), which fits with this reconstruction. The vast majority of the steatite tablets reported by Vats come from his Strata V and VI which, again accepting his below-surface measurements and correlating them with our excavations, would be Period 3B and the first part of Period 3C of our periodization. This distribution of inscribed materials was confirmed by our excavations in this area.

TABLE 2 – *Mound F, Trench I depths (Vats 1940: 80-110), elevations (calculated using modern surface elevation of 164.00 m) and significant archaeological observations*

Vats Stratum	Depths below surface		Elevations above sea level	
	Top	Base	Top	Base
I	Surface	4' (1.22 m)	164.00 m	162.78 m
II	4' (1.22 m)	5'6" (1.68 m)	162.78 m	162.32 m
III	5'6" (1.68 m)	8'7" (2.62 m)	162.32 m	161.38 m
IV	8'6" (2.59 m)	11' (3.35 m)	161.41 m	160.65 m
V	11' (3.35 m)	14'6" (4.42 m)	160.65 m	159.58 m

VI	14'7" (4.45 m)	18'6" (5.64 m)	159.55 m	158.36 m
VII	18'6" (?) (5.64 m)	22' (?) (6.71 m)	158.36 m	157.29 m
VIII	22' (?) (6.71 m)	25'6" (7.77 m)	157.29 m	156.23 m
End		30' or 34'6" (9.14 or 10.52 m)		154.86 m or 153.48 m
Vats			HARP- H96	
brick-robber tunnels to 12' (160.34 m elevation)			brick-robber debris to c. 160.5 m	
impressed PGB at 12' (160.34 m elevation)			earliest PBGs at c. 160.4 m	
earliest steatite tablet at 19' (158.21 m elevation)			earliest tablet at c. 159 m	
earliest seal at 20'9" (157.68 m elevation)			hearth [101] at c. 157.10 m	
water table at c. 30'-34'6" (154.86-153.48 m)			water table at 156.54 m elevation	

Perhaps it is possible to narrow the chronology of Trench I even more. As noted above (Table 1), a calibrated date of 2328 BC was obtained from hearth [101] near the bottom of the street deposits identified in our excavations. Another date with calibration intercepts at 2272, 2258, and 2204 BC comes from street deposits c. 1.5 m above the hearth. Accepting the depth measurements of Vats, the first date would be in the upper part of his Stratum VIII, and the second would be in his Stratum VI (Table 2). If we take these and our estimate for the duration of Period 3B (2450 to 2200 BC) at face value, this would mean that most of the intact deposits of Trench I (Vats Strata V, VI, and VII) belong to the second half of Period 3B and to the earliest part of Period 3C. And even if most of the lower deposits of Mound F, Trench I, actually cover all of our Period 3B and the earliest cultural layers extend into Period 3A, the inscribed material is concentrated in the upper part of that sequence (Vats Strata VI and V, as noted), with the lowest steatite tablet only in what would be the upper few centimetres of Stratum VII. Thus, on the evidence of Trench I alone, we must reject the notion that the steatite tablets ('tiny steatite seals') come from the early part of the Harappan Phase at Harappa. They seem instead to belong primarily to the second half of HARP Period 3B and the early part of Period 3C.

The Archaeology of Mound E, East Side (HARP Trench 11)

Considerable further information on the tablets and their dating comes from our excavations on the east side of Mound E. In *South Asian Archaeology 1995* we gave a first account of the excavations in HARP Trenches 10W and 11 in this area of the site (Meadow & Kenoyer 1997). Trench 10W included two rooms ([92] and [93]), each with three hearths, situated on the west side of a major N-S street just below the modern surface of the site. Nine radiocarbon dates have been obtained from charcoal collected in these hearths (A-7522, QL-4732, A-7523, QL-4733, A-7524, A-7528, QL-4734, A-7529, QL-4730: Meadow & Kenoyer 1997: Table 1). Seven of the nine determinations calibrate between 2450 BC and 2207 BC; of the other two, one is much earlier and one much later. Excluding the outliers and averaging the seven (using Calib. 3.0.3c) provides a calibrated date intercept of 2327 BC (3869 ± 13 bp with a 97% probability that the average date falls between 2403 and 2282 BC). Thus, based on radiocarbon dates alone, we can place these rooms in our Period 3B, an attribution that is confirmed by the nature of the ceramic corpus and by the stratigraphy of the area.

More than 10 m to the West of the Room [92]/[93] complex is the N-S perimeter wall (Wall [250]) on the east side of Mound E. Originally located in 1992, this c. 6 m wide mud-brick wall is thought to have been first constructed in Period 3A at or near the beginning of the Harappan Phase. A deep sounding dug in 1995 against the outer face of the wall revealed the existence of at least three metres of Period 3A and 3B street deposits. Below this were found about two metres of late Period 2/early Period 3A occupational debris and below that, natural sediment (Meadow & Kenoyer 1997: fig. 6; see end note 2). During the 1996 and 1997 seasons excavations in the area between Rooms [92]/[93] and Wall [250] were extended over a zone about 30 m long (Fig. 2). This work confirmed the sequence we suggested in our 1995 article (Meadow & Kenoyer 1997).

Sometime in the second half of Period 3B (perhaps about 2300 or 2350 BC), Wall [250] was repaired and the domestic structures to the East were levelled (including the Room [92]/[93] complex). This left a wide open space outside of the perimeter wall. Over time (probably a few years if not less), debris – including hundreds of baked bricks – was dumped over this wall. Subsequently a low 80 cm wide mud-brick curtain wall

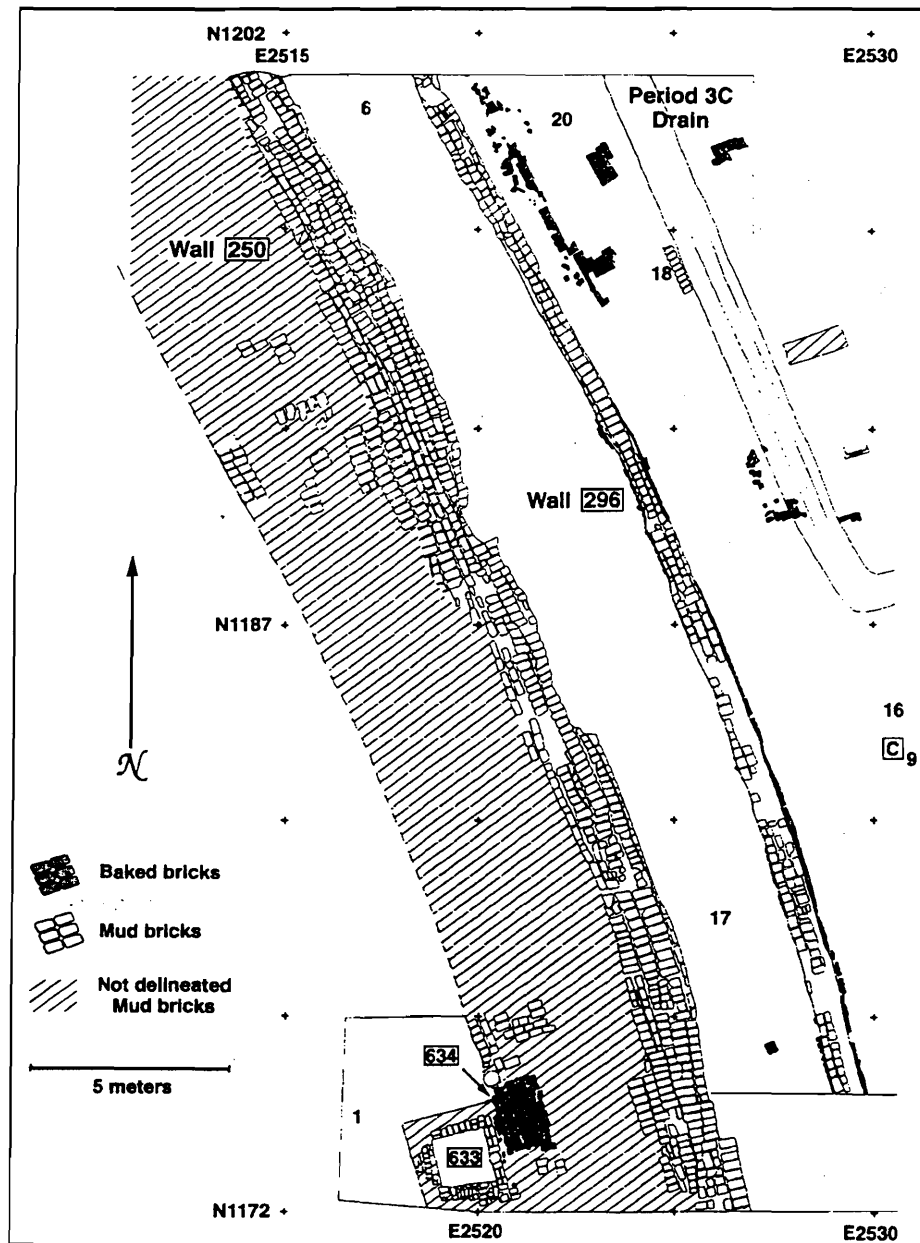


Fig. 2 –Harappa 1997: Mound E, HARP Trench 11: perimeter walls, associated structures, and locations of inscribed pieces illustrated in Fig. 4.

(Wall [296]) was constructed at least partially against this debris; in places some of the baked bricks that had been dumped over Wall [250] were used for a footing for the new curtain wall. The area between the two walls was then filled with debris to the top of the curtain wall, thus forming a four-metre wide revetment along the whole outside face of the perimeter wall. Shortly thereafter, debris also began to be dumped over the outside of this revetment where it was deposited in clumps merging with one another and sloping down to the East over the previously levelled area. Debris also was deposited on top of the revetment, and this merged with the material on the slope. Eventually compact street-like deposits resulted, some of which – especially above the revetment – included a significant component of decayed baked brick.

Through the remainder of the Harappan Phase, debris probably continued to be dumped over the perimeter wall, although there is little direct evidence for this as subsequent erosion would have removed such deposits. It is clear from the nature of the sediments inside the perimeter wall, however, that Wall [250] once stood at least two metres higher than it is preserved today. Horizontally laid street deposits continue well above the elevation of the modern surface level of the wall which, being constructed of mud-bricks, is more susceptible to erosion than the compacted street debris that lay against it on the inside (Meadow & Kenoyer 1997: fig. 6). These street deposits all contain pointed base goblets and thus can be assigned to our Period 3C. The uppermost debris preserved above the revetment on the outside of Wall [250] also contains a few pointed bases and scored bodies and thus can probably be attributed to the early part of Period 3C. All materials below this, however, including the sloping layers against Wall [296] and the fill between Walls [250] and [296] are devoid of Indus goblets and can be dated to the second half of Period 3B.

At the south end of the excavated area, a small mud-brick structure was found projecting out of Wall [250] into the street that ran along the inside of the perimeter wall. Just East of this room [633] was found a baked-brick platform [634] that had been cut into or laid on top of the perimeter wall. How these features were used in relation to Wall [250] and, more especially, what the perimeter wall looked like when these features were in use is not clear because the features were found in a heavily eroded area covered by debris that had obviously been deposited fairly recently (to judge by the pieces of plastic in the fill). The lower fill of Room [633] and the adjoining

street deposits were intact, however. The former yielded two small plain vessels, while the latter were found to contain a ceramic assemblage that is much like that which had been dumped outside Wall [296].

Finally, sometime during Period 3C (probably late in that period), a massive drainage ditch was dug deep into the underlying Period 3B strata from a level that was above today's surface. The lower fill of this drain contained many pointed base goblets, some complete or nearly so, while the upper fill included material washed in from the deposits through which the drain had been cut. There are also places in Trench 11 that have been severely affected by erosion or brick-robbing. These too were found to contain mixed debris, which we removed separately from the intact deposits through or into which the erosion had cut.

Inscribed Materials from Mound E, East Side (HARP Trench 11)

As noted in our previous report, the Trench 11 area has yielded a large number of inscribed pieces. To date we have tabulated 5 intaglio seals, 73 moulded tablets (terracotta and faience), 133 incised steatite tablets, and many inscribed sherds. The majority of this material comes from the sloping debris layers to the East of the curtain wall [296], some is from the deposits on top of the revetment formed by the fill between Walls [250] and [296], and some was found in the fill between those two walls. A few pieces come from earlier levels, i.e., in the upper street deposits in the deep sounding East of Wall [250], and some were recovered from later or mixed deposits across the area. Among the moulded tablets there are many duplicates (made from the same set of moulds) while among the steatite tablets there are many copies (same signs but each tablet inscribed individually). Although both duplicates and copies are well known from the earlier excavations at Harappa (e.g., Vats 1940; Mahadevan 1977; Parpola 1994), those from the Trench 11 area provide some new insights into chronology and possibly function.

We have already reported and illustrated an important set of three images and one group of signs impressed into terracotta that can occur in different combinations on two-, three-, and four-sided tablets (Meadow & Kenoyer 1997: fig. 9). The three images were described by Vats (1940: 129, pl. XCIII 310) on a tablet from Mound F, Trench VI, Stratum III, 6.5 feet

below surface. We have now recovered 13 tablets with two or more of these images/sign group – 12 from Trench 11 and one from Trench 27 (Mound ET). Five four-sided specimens and one two-sided example that were found during the 1997 season are illustrated here as Figure 3.1-6. Of these, five come from Period 3B debris East of Wall [250], while one (Fig. 3.5) comes from heavily disturbed deposits West of that wall. Given that most of these tablets come from well-defined Period 3B deposits, we can with some confidence assign their manufacture to that period and, as noted, probably to the second half of the period. Their appearance also in contexts with pointed-base goblets (in Trench 11 West and Trench 25) confirms that what appeared upon excavation to be disturbed or mixed deposits were indeed that.

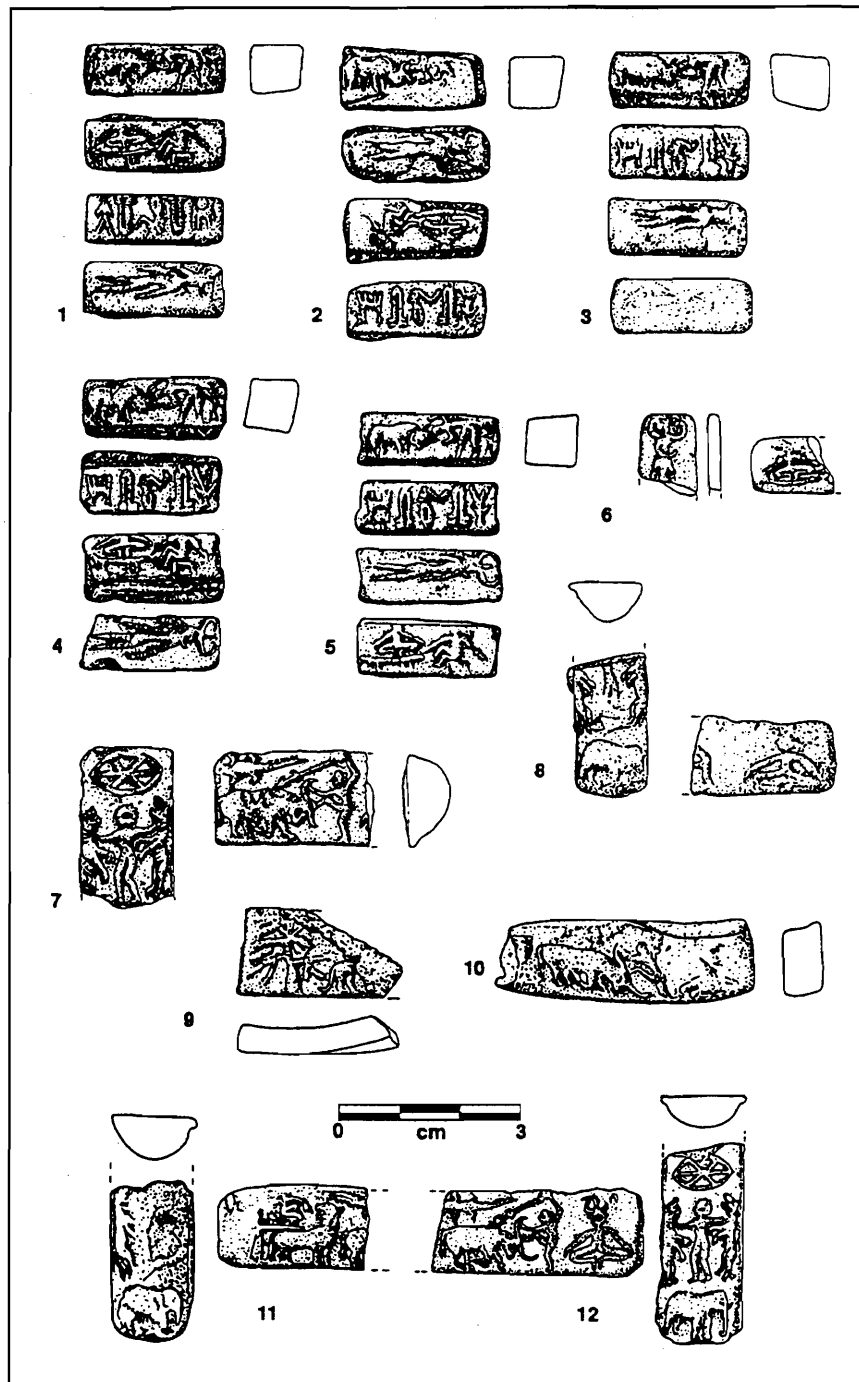
The same conclusions can be reached with another tablet set, this time of two images which, because of their elaborate nature, we originally thought could be attributed to Period 3C (Kenoyer & Meadow 1996; Meadow & Kenoyer 1997). These images have been found only on one- or two-sided terracotta tablets, four of which from the 1997 season are illustrated as Figure 3.7-10 and two of which from previous seasons are illustrated as Figure 3.11-12. As with the previous set, we have already reported on the iconography of these pieces and illustrated an excellent example from mixed fill in Mound ET, Trench 10E (Kenoyer & Meadow 1996: fig. 6.1; Meadow & Kenoyer 1997: fig. 9.8; here Fig. 3.11). On the convex side of this tablet is what appears to be a woman holding a tiger by the neck at the end of each outstretched arm. Above this 'combat' scene is the Harappan script 'wheel' motif, while below it is an elephant. On the flat side is what appears to be a man spearing a water buffalo and a second horned and banded individual sitting in the 'yogic' position. Above the water buffalo is a gharial, but with its head missing because of where the tablet is broken. A second example was found during the 1996 season in what was thought to be Period 3C or mixed context in another area of Mound ET (Trench 35; here Fig. 3.12). That piece is a duplicate on both sides of the one found by Vats (1940: 59, pl. XCIII 308) in the northeast corner of Trench IV, Mound F in Stratum IV, 4-7 feet below the surface. It depicts on the convex side the woman in combat with tigers above the elephant and on the flat side a man sitting in a tree above a tiger. In the case of the piece we found in 1996, however, there is enough of the tablet preserved on the right side of the scene on the flat side to show the head of a gharial and the hind quarters of

another animal to the right of the tiger (Fig. 3.12). We had previously suggested that this was a different scene than that of the man spearing the water buffalo (Kenoyer & Meadow 1996: 12) but the tablet from Mound ET and the twisted tablet found this season (Fig. 3.10) clearly indicate that these are two halves of the same scene. This scene would have begun on the left with the man in the tree above the tiger, followed by the man spearing the water buffalo below the gharial and then by the individual in 'yogic' position. On the convex side, however, because of the way the tablets were impressed, we have only the scene as described above and still do not know what appeared above the 'wheel' sign.

Vats (1940: 105, pl. XCI 248) reports and illustrates a scene of man in tree above tiger, but incised on a 'bossless, rectangular seal of black steatite' from Stratum VI South of the 'deep digging' in Trench I of Mound F at a depth of 14 feet 9 inches. This provenience, if we can take the depths below surface at face value, suggests a Period 3B date, which can now be confirmed by the four specimens from Trench 11. The best preserved of these (Fig. 3.7) was found just above the surface upon which curtain Wall [296] was constructed and thus comes either from the rubble that was tossed over Wall [250] before Wall [296] was built or from the fill that was dumped between the two walls. The other three specimens are more worn (Fig. 3.8-10) and come from terminal Period 3B layers just above or adjacent to the top of curtain Wall [296]. Of these, one is a duplicate of the previously noted specimen while the other two are both flat twisted terracotta tablets with only one side decipherable. In one case the man in tree above tiger scene is presented (Fig. 3.9); in the other the man spearing water buffalo scene can be made out (Fig. 3.10).

Both of the previous groups of terracotta tablets were formed from sets of master moulds that could produce multiple duplicates over a period of time. How long such a period might have been is not known, although one can suspect that it was not more than half a century. Also not known are how long any given tablet may have remained in use and, of course, what they were used for. Although the evidence needs to be analysed in greater detail, we can now suggest that, at Harappa, elaborate iconographic scenes began to be moulded onto terracotta tablets at about the middle of Period 3B and that they continued to be made into Period 3C, although for how long is not clear.

In addition to the duplicates among the moulded terracotta and faience



tablets, there are a number of copies among the incised steatite tablets from the Trench 11 area. In these cases the same signs or sign sequences were engraved (copied) into different blocks of steatite which were subsequently fired. We have already reported some copies from this area (Meadow & Kenoyer 1997) including an example of two tablets being mirror images of each other. Here we present a set of 22 tablets with the same inscriptions (Fig. 4). Each tablet is three-sided with the inscription on each side comprising a single more complex sign accompanied by three or four simple strokes. All of the tablets were found in the debris that was dumped after the curtain wall [296] was constructed and after the area between that wall and perimeter wall [250] had been filled in, but before any Period 3C material was deposited in the area. Sixteen of the tablets were found together in a clump (Fig. 2: designation 'C' and Fig. 4), two others were found nearby (Fig. 2: nos. 9 and 16) and may originally have come from the same cluster, and the remaining four were found scattered in the area at the approximate locations noted in Fig. 2 (nos. 6, 17, 18, 20).

Fig. 3 – Harappa 1995-1997: Mounds E and ET: molded terracotta tablets.

Mound E, HARP Trench 11:

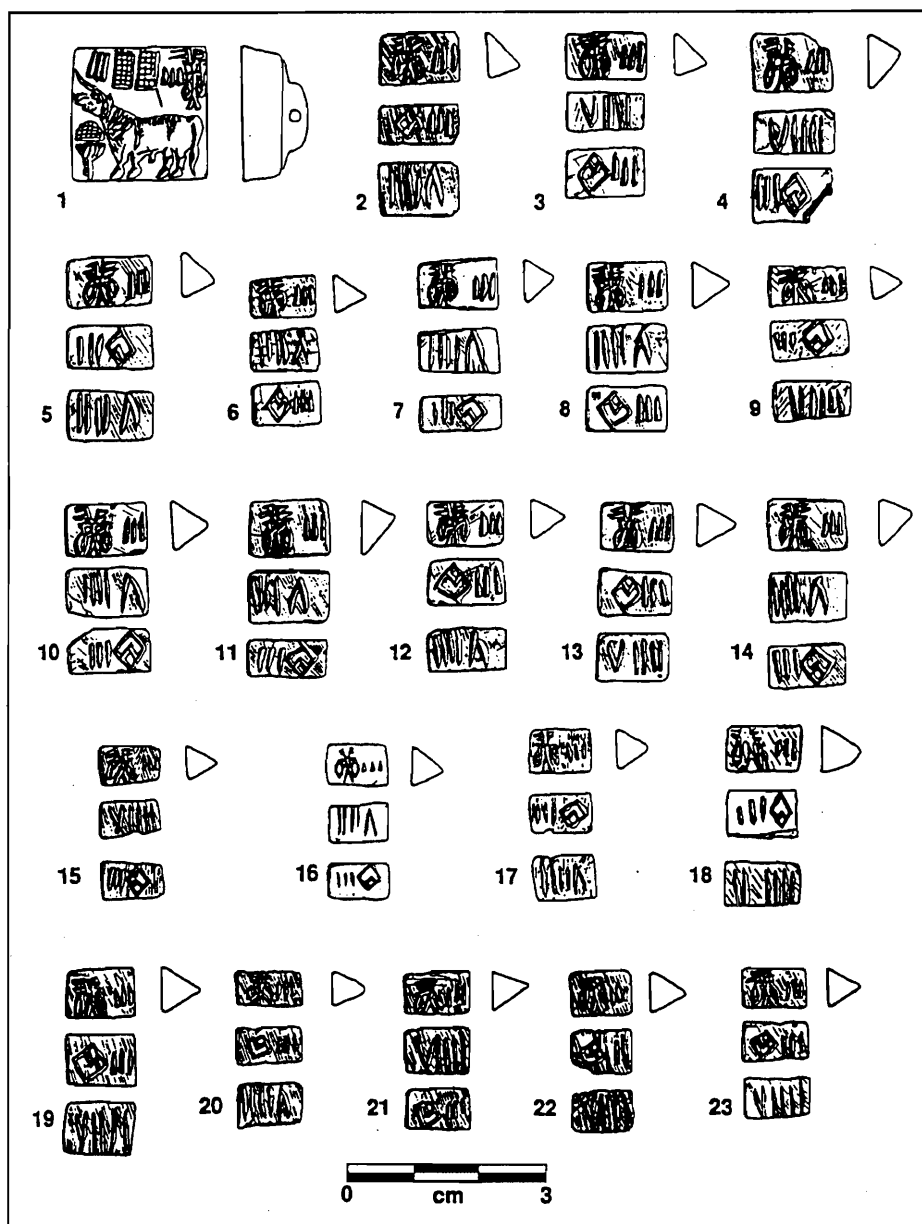
1. H97-3343/8029-28
2. H97-3346/8029-31
3. H97-3352/8029-32
4. H97-3301/8035-01
5. H97-3349/8102-41
6. H97-3345/8029-30
7. H97-3405/8250-02
8. H97-3429/8010-76
9. H97-3428/8029-71
10. H97-3356/8066-01

Mound ET, Trench 35:

11. H96-3119/6252-01

Mound ET, Trench 10E:

12. H95-2486/4651-01



As is evident in Fig. 4, while each of the tablets bears the same set of inscriptions, they do not always occur in the same sequence or orientation as one flips the tablet from one face to the next. This fact demonstrates that the stone engravers did not turn the tablet the same way each time when making the inscriptions. It also indicates that the sequence of character sets and the relation of these sets to each other was not an important feature of these tablets. Microscopic examination by one of us (JMK) of fine-grain moulds made from these tablets has further indicated that while they were all incised with strokes made in the same direction and in the same sequence for each character, they were incised by at least three different stone engravers. In Fig. 4, tablets nos. 2-7 make up one group, nos. 8-16 a second group, and nos. 17-23 a third group. All three groups are represented in the tablet cluster, and all three are represented by single tablets as well – unless tablets nos. 9 and nos. 16 also originally came from the cluster (see above), in which case Group 2 would be represented only in the cluster. Different explanations for these phenomena present themselves. All three groups of tablets may have been made at one time in one workshop by

Fig. 4 – Harappa 1995-7: Mound E, HARP Trench 11: steatite seal and incised steatite tablets.

Steatite seal

1. H96-2796/6876-01

Steatite tablet manufacturing group 1:

2. H97-3304/8040-01

3. H97-3305/8040-02

4. H97-3312/8040-05

5. H97-3314/8040-07

6. H97-3290/8010-03

7. H97-3315/8040-08

Steatite tablet manufacturing group 2:

8. H97-3318/8040-12

9. H97-3333/8038-01

10. H97-3317/8040-11

11. H97-3319/8040-13

12. H97-3316/8040-09

13. H97-3313/8040-06

14. H97-3306/8040-03

15. H97-3307/8040-04

16. H96-3046/6951-04

Steatite tablet manufacturing group 3:

17. H96-3125/6937-16

18. H95-2613/6560-01

19. H97-3311/8040-10

20. H97-3341/8039-10

21. H97-3320/8040-14

22. H97-3322/8040-16

23. H97-3321/8040-15

three different carvers, or they may have been made at three or more times in one or more workshops by three different carvers. If the latter, at least 16 of them would have had to have been accumulated in one place by an individual or institution.

Smaller clusters of incised steatite tablets as well as multiple copies from the same general area have been found previously at Harappa, notably in Trench I of Mound F (Vats 1940). What is particularly important about the specimens from Trench 11, however, is that because of the nature of the excavation it is possible to show that they all come from contemporary or nearly contemporary stratigraphic units, and it seems likely that although made by different hands they derived from a single source. But whatever value or use they may have had when they were made had clearly been lost by the time they were deposited together with other debris on the outside of the Mound E perimeter wall. Of great interest in this regard is a unicorn seal (Fig. 4.1) that was found inside the perimeter wall in the approximate location marked in Fig. 2 (no. 1). The last two signs of this seal are the same as those on one side of the 22 tablets (taking the three strokes as a single sign). These are preceded by three additional signs (reading left to right on the seal as opposed to on the tablets which we assume are meant to be read right to left because they were not meant to be impressed – see endnote 3). It is rare to find the same sign sequence on a seal and on tablets, and the situation is that much more compelling because the street deposits from which the seal comes contain the same ceramic corpus as the dump deposits from which the tablets come, suggesting that both were originally formed at about the same time.

It is tempting to think that the evident loss of utility and subsequent discard of the tablets is related to the ‘death’ of the seal. Seals are almost always found in trash or street deposits (and never yet in a grave) indicating that they were either lost or intentionally discarded, the latter seeming the more likely in most instances. The end of the utility of a seal must relate to some life event of its owner, whether change of status, or death, or the passing of an amount of time during which the seal was considered current. A related consideration is that apparently neither seals nor tablets could be used by just anyone or for any length of time because otherwise they would not have fallen out of circulation. Thus the use of seals – and of tablets – was possible only if they were known to be current. Once they were no longer current, they were discarded. This would help explain why

a group of 16 (or 18) tablets with the same inscriptions, kept together perhaps in a cloth or leather pouch, could have been deposited with other trash outside of the perimeter wall of Mound E.

Conclusion

The incised steatite tablets from Harappa can no longer be viewed as among the first objects specifically shaped to bear the Indus script. Instead they should be seen as a means peculiar to Harappa of employing the script for a specialized purpose at a particular time in the history of the city. Their production seems to be more or less coincident with the moulding of iconographic scenes onto terracotta tablets beginning perhaps in the middle of HARP Period 3B or c. 2350-2300 BC. This was a period of significant 'urban renewal' at Harappa, with a major reconstruction of the east side of Mound E and perhaps also with the construction of the 'granary' on Mound F (based on work carried out in 1997). This is also a time coincident with the rise and florescence of the Akkadian Empire in Mesopotamia. What might be considered in the future is whether this timing is mere coincidence or is a reflection in some way of the interaction between the two regions that is attested in the Mesopotamian texts and by Indus materials in Mesopotamian contexts.

As noted in our article in *South Asian Archaeology 1995*, it is essential that we no longer deal with the Indus Civilization as a phenomenon undifferentiated in time and space. The archaeological remains that we know from Mohenjo-daro, Harappa, and Dholavira come primarily from the second half of the five to seven hundred year Harappan Phase. Our understanding of the first half is based only on small soundings at the major sites, and even at smaller sites such as Chanhudaro and Nausharo, the extent and depth of later material has made wide exposures of earlier levels almost impossible. Thus we know very little of how the Indus Civilization initially developed at any site. With the discovery of the existence of the whole sequence of occupation at Harappa on the north side of Mound AB in an area so eroded that exposure of significant exposures of all periods is possible (Kenoyer & Meadow, this volume), we have some hope of addressing this issue at Harappa.

Endnotes

1) Our 1996 excavations in Trench I of Mound F have provided important information about the rise of the water table at Harappa, which we found to be at an elevation of 154.86 m amsl in April. Accepting the depth measurements of Vats (1940), this would be within his Stratum VIII (see Table 2). Vats himself reports that his excavators encountered water at 30 feet below surface (p. 80) or at 10 feet below the drain of Stratum VIII (p. 97), which would actually be a depth of 34 feet, 6 inches using the depth of the drain as reported (Vats 1940: 92). In the one case the water table would have risen by 1.68 m since his excavations, in the other case by 3.06 m. In any event, it is clear that cultural deposits at Harappa, in areas of the site that were originally lower lying, are now under water. We were thus doubly fortunate to find that the earliest occupation at the site was actually at a relatively high elevation, probably on an old levee or terrace of the river (Kenoyer & Meadow, this volume).

2) The designations of Periods at Harappa have been adjusted a bit since the discovery of the 'pre-Kot-Dijian' deposits (now Period 1) on the north side of Mound AB (see Kenoyer & Meadow, this volume). Period 2, previously considered as the 'Early Harappan/Harappan Transitional', is now used to refer to the 'Kot Dijian' manifestation at Harappa. For the time being, material thought to be transitional to the Harappan phenomenon at Harappa is being designated as Period 2/3A.

3) The question of the direction in which the Indus script is to be read is an important one and the many previous writings on the subject seem to us not always to have been very logically considered ~~one~~ (see recent reviews in Parpola 1994 and Possehl 1996). To our mind it is essential to take careful note of the production process and the medium on which the writing occurs. For example, steatite intaglio seals were manufactured by craftsmen who may themselves not have been literate. Thus, the direction in which script was engraved on seals need have nothing to do with the way they were to be read. This is particularly the case if the script on the seals was meant to read correctly in impression. If the seal maker was incising the signs in the *order* in which they were to be read, he or she would actually be engraving the signs in mirror image *and* in the opposite direction from

which they were to be read. We believe that some confusion has come from the fact that seals – and sometimes other incised pieces as well – are often pictured, even when published for the first time, as impressions and not as they actually are. We consider this to be a questionable practice. It foists an interpretation onto the material – that it was the impressions that were meant to be read – and causes logical difficulties such as those referred to above, particularly as it is not always noted in captions whether the object is being displayed as an impression or as it actually is. This practice of depicting impressions, even for the flat seals of the Indus world, probably derives from the necessity of publishing impressions of cylinder seals the scenes on which cannot otherwise be completely presented. We urge that, in the future, all inscribed pieces from the Indus world initially be published as they actually occur, and that it be the task of subsequent researchers to make whatever transformations they feel necessary to make. And whatever transformations are made need to be carefully noted for each and every illustration. This will bring the publication of inscribed pieces into conformance with the standard archaeological practice of depicting the object as it exists. The only exception would be for cylinder seals which, as noted, need to be presented in impression, although here too a view of the seal itself should be presented as well as a view of the impression.

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Acknowledgments

Excavations by the Harappa Archaeological Research Project are conducted in collaboration with the Department of Archaeology and Museums, Government of Pakistan. We wish particularly to thank Dr. M. Rafique Mughal (Director-General 1996), Mr. Niaz Rasool (Director-General 1997), Mr. M. Bahadur Khan (Curator, Harappa Museum, and Field Representative 1996), and Mr. Habib ullah Nasir (Assistant Curator, Harappa Museum, and Field Representative 1997) for their assistance during the 1996 and 1997 field seasons.

Support for these field seasons was provided by the National Endowment for the Humanities (1996 season only), the National Geographic Society, the American School of Prehistoric Research, the Peabody Museum of Archaeology and Ethnology, Harvard University, the University of Wisconsin, New York University, the Smithsonian Institution, the Kress Foundation, and private donors.

In addition to the efforts of our Pakistani workmen, we wish to acknowledge the contributions made by Ms. Sharri Clark, Ms. Maria Jaffery, Ms. Usma Rizvi, Ms. Naila Shabnum, and Mr. Robert Simpkins as excavation supervisors in the areas discussed in this paper.